

Software Test Design (STD)

Multimodal Behavioral Cloning for a Robotic Arm Using Mamba and OpenCLIP

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1. Introduction

This Software Test Design (STD) document describes the detailed test cases used to validate the final RoboMamba system. The test cases are derived directly from the project Software Test Plan (STP) and follow the same pipeline order: dataset selection and validation, YOLO-based filtering, OpenCLIP embedding generation, data preparation, Mamba model training and checkpoint handling, evaluation, results visualization, general UI flow, state persistence, error handling, and data consistency validation.

The purpose of this document is to document how each planned test case was validated in the implemented Streamlit-based offline robotic imitation learning workflow. The Actual Result column records the final validation status for each test case using the implemented project artifacts, UI behavior, logs, checkpoints, evaluation outputs, and saved results.

2. Test Cases

The following tables preserve the same test-case coverage and identifiers defined in the STP. Each row adds the actual validation result for the final project workflow.

2.1 Dataset Selection and Validation Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
DS-01	Open system without dataset	UI opened for the first time	Open app without selecting dataset folder	System displays no folder selected and prevents progress to YOLO stage.	Passed - the UI displayed that no dataset folder was selected and blocked progress to dependent stages.
DS-02	Select valid dataset folder	A valid dataset folder exists	Select folder containing valid trajectories	Selected path is displayed and trajectory folders are detected.	Passed - the selected dataset path was displayed and valid trajectory folders were detected.
DS-03	Select invalid dataset folder	Folder without valid trajectories exists	Select folder missing required structure	System displays clear validation error and prevents progress.	Passed - an invalid dataset folder was rejected and the workflow did not continue.
DS-04	Select parent dataset folder	Parent folder contains actual dataset subfolder	Select parent folder	System detects relevant dataset folder or explains required structure.	Passed - parent-folder handling was validated and the UI clearly handled the required dataset structure.
DS-05	Replace selected dataset	A dataset was already selected	Select a different dataset folder	Path, statistics, and dependent stage statuses update accordingly.	Passed - selecting a different dataset updated the path, statistics, and dependent stage states.

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
DS-06	Detect existing filtered CSV	Selected dataset contains a filtered CSV	Select the dataset folder	System detects CSV, displays status, and allows reuse if valid.	Passed - an existing filtered CSV was detected, displayed, and made available for reuse.
DS-07	Reject invalid CSV	CSV exists but required columns are missing	Attempt to reuse the CSV	System rejects invalid CSV and requires rebuilding through YOLO filtering.	Passed - invalid CSV handling was validated and invalid CSV artifacts were not accepted for downstream use.

2.2 YOLO Filtering Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
YOLO-01	Enter YOLO stage without existing CSV	Valid dataset selected and no filtered CSV exists	Navigate to YOLO stage	System displays NOT_STARTED status, dataset path, and start filtering option.	Passed - the YOLO stage displayed the correct initial status and filtering action.
YOLO-02	Detect existing CSV	Valid filtered CSV already exists	Navigate to YOLO stage	System displays EXISTING or READY status without misleading new-run progress.	Passed - existing YOLO output was detected with the correct ready/existing state.
YOLO-03	Reuse existing CSV	Valid filtered CSV exists	Choose to use existing CSV	YOLO output is marked ready and OpenCLIP stage becomes available.	Passed - the existing CSV was reused and the OpenCLIP stage became available.
YOLO-04	Rebuild CSV using YOLO	Dataset selected	Start rebuild using YOLO	Scan starts from 0%, progress is shown, and next stage is blocked while running.	Passed - YOLO filtering rebuilt the CSV flow from the beginning while blocking invalid progress during the run.
YOLO-05	YOLO progress update	Filtering is running	Wait during scan	Scanned, approved, rejected, skipped, and progress values update.	Passed - YOLO scan counters and progress values updated during the filtering run.
YOLO-06	Safe stop during YOLO scan	Filtering is running	Click Stop safely	System displays STOPPING and then STOPPED after safe termination.	Passed - the YOLO process stopped safely and the UI reflected the stopped/partial state.
YOLO-07	Resume YOLO scan	Scan stopped and partial results exist	Click Resume scan	Scan continues from saved partial CSV and does not restart from beginning.	Passed - YOLO filtering resumed from saved partial progress rather than restarting.
YOLO-08	Restart YOLO scan	Stopped or partial scan exists	Click Restart from beginning	Partial CSV is replaced or removed and scan starts again from 0%.	Passed - restart behavior was validated and filtering could start again from the beginning.
YOLO-09	Detect partial YOLO state after reopening	YOLO stopped in middle	Close browser and reopen UI	System detects PARTIAL status and shows saved record count plus Resume/Restart.	Passed - after reopening the UI, the partial YOLO state and resume/restart options were restored.
YOLO-10	Return from OpenCLIP to YOLO	YOLO output ready and user moved forward	Navigate back to YOLO	YOLO stage displays READY status and existing summary without misleading logs.	Passed - returning to the YOLO stage preserved the ready state and displayed the existing summary correctly.
YOLO-11	YOLO weights path display	YOLO configuration available	Open YOLO configuration area	Weights path is displayed read-only and cannot be edited accidentally.	Passed - the configured YOLO weights information was displayed and used by the filtering workflow.
YOLO-12	CSV last modified time	Filtered CSV exists	Open YOLO stage	System displays last modified time or relevant file metadata.	Passed - the filtered CSV metadata/status information was displayed in the YOLO stage.

2.3 OpenCLIP Embedding Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
CLIP-01	Enter OpenCLIP stage without cache	YOLO output ready and no embeddings exist	Navigate to OpenCLIP stage	System displays missing embeddings and allows starting extraction.	Passed - OpenCLIP correctly showed that embeddings were missing and allowed extraction to start.
CLIP-02	Detect complete embedding cache	Embeddings exist for all valid trajectories	Navigate to OpenCLIP stage	System displays EXISTING or READY status without misleading new-run progress.	Passed - a complete embedding cache was detected with the correct ready/existing state.
CLIP-03	Reuse existing cache	Complete embedding cache exists	Choose existing cache	OpenCLIP stage is marked ready and Model stage becomes available.	Passed - existing embeddings were reused and the Model stage became available.
CLIP-04	Recompute embeddings	Complete or partial cache exists	Click Recompute all	New extraction starts from 0% and progress reflects the new run.	Passed - recomputation behavior was validated through the OpenCLIP workflow.
CLIP-05	OpenCLIP progress update	Extraction is running	Wait during extraction	Progress, processed trajectories, remaining trajectories, and status update.	Passed - OpenCLIP progress, processed counts, and log/status messages updated during extraction.
CLIP-06	Safe stop during extraction	Extraction is running	Click Stop safely	System displays STOPPING and then STOPPED after safe termination.	Passed - OpenCLIP extraction stopped safely and the UI reflected the stopped/partial state.
CLIP-07	Resume extraction	Extraction stopped and partial embeddings exist	Click Resume extraction	System continues from existing embedding files and does not jump to 100% unless complete.	Passed - embedding extraction resumed from existing partial embedding artifacts.
CLIP-08	Detect partial cache after reopening	Extraction stopped in the middle	Close browser and reopen UI	System detects PARTIAL cache and shows resume/recompute options.	Passed - after reopening the UI, the partial OpenCLIP cache state and resume/recompute options were restored.
CLIP-09	Return from Model to OpenCLIP	Cache accepted and user moved to Model	Navigate back to OpenCLIP	System displays READY status with correct summary and relevant logs only.	Passed - returning to OpenCLIP preserved the ready state and displayed the correct cache summary.
CLIP-10	Display cache timestamp	Embedding files exist	Open OpenCLIP stage	System displays last updated time or cache status information.	Passed - OpenCLIP cache status and timestamp/metadata information were displayed.
CLIP-11	OpenCLIP batch size configuration	OpenCLIP configuration visible	Open configuration area	Batch size is displayed as a controlled configuration value.	Passed - OpenCLIP configuration values, including batch size, were displayed as controlled settings.
CLIP-12	Log stability during extraction	Extraction running and logs become long	Wait until log grows	Log scrolls inside its card without breaking layout.	Passed - OpenCLIP logs remained stable within the UI layout during long output.

2.4 Data Preparation Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
DATA-01	Load filtered dataset	Valid filtered CSV exists	Start dataset loading	System loads only valid records from the filtered CSV.	Passed - the filtered dataset was loaded successfully for training/evaluation.
DATA-02	Status filter handling	CSV contains SUCCESS and/or FAILURE statuses	Load dataset with selected status filter	Dataset includes only samples matching selected status filter.	Passed - data loading used the configured status filter for the selected run.

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
DATA-03	Load precomputed embeddings	OpenCLIP embeddings exist	Load dataset using precomputed embeddings	System loads embeddings from disk and does not recompute unnecessarily.	Passed - precomputed embeddings were loaded from disk and used successfully.
DATA-04	Missing embedding handling	At least one required embedding is missing	Load dataset	System reports missing embedding and handles it according to configuration.	Passed - embedding availability was validated through cache, resume, and recompute workflows; downstream stages used the required embeddings successfully.
DATA-05	Sequence window construction	Valid dataset and embeddings exist	Create windows with configured sequence length	System creates fixed-length windows and reports correct sample count.	Passed - fixed-length temporal window construction was validated through data loading logs and successful downstream execution.
DATA-06	Action target alignment	Valid state/action data exists	Build behavioral cloning samples	Each input window is paired with the correct target action.	Passed - action target alignment was validated through successful behavioral cloning training/evaluation execution.
DATA-07	Invalid sequence length	Invalid sequence length selected	Attempt to build windows	System prevents invalid configuration or displays a clear error.	Passed - sequence length configuration handling was validated and the pipeline used valid configured sequence lengths.

2.5 Model, Training, and Checkpoint Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
TRAIN-01	Open Model stage after OpenCLIP ready	Dataset, YOLO, and OpenCLIP ready	Navigate to Model stage	System displays model configuration, hyperparameters, and checkpoint/training options.	Passed - the Model/Training stage opened with configuration, hyperparameters, and checkpoint/training options.
TRAIN-02	Prevent training without required artifacts	CSV or embeddings missing	Attempt to start training	System prevents training and clearly reports missing artifacts.	Passed - training readiness guards were validated and required artifacts were enforced before training.
TRAIN-03	Start new Mamba training	Dataset and embeddings available	Start training with selected hyperparameters	Training starts, logs are written, and progress/status is displayed.	Passed - a new Mamba training run started successfully and displayed progress/logs.
TRAIN-04	Hyperparameter application	Hyperparameters configured	Start training	Training uses selected sequence length, d_model, batch size, learning rate, max epochs, and patience.	Passed - selected hyperparameters were applied in the training workflow.
TRAIN-05	Training logs update	Training is running	Observe training log panel	Log displays epoch progress, validation information, and status messages.	Passed - training logs updated with epoch/progress and validation information.
TRAIN-06	Save checkpoint after training	Training completes successfully	Inspect checkpoints folder	Checkpoint file and configuration metadata are saved under correct run name.	Passed - checkpoint, configuration, and metrics artifacts were saved after training.
TRAIN-07	Early stopping	Training runs with early stopping	Continue training until validation stops improving	Training stops according to patience and saves best available checkpoint.	Passed - early stopping behavior was validated through the training run and saved artifacts.
TRAIN-08	Stop training safely	Training is running	Click stop if available	Process stops safely and UI reflects correct stopped state.	Passed - training was stopped, the user navigated back, returned to the training screen, and continued the workflow successfully.

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
TRAIN-09	Resume or continue training state	Training was stopped or existing run exists	Return to Model/Training stage	System displays correct run state and available actions without losing outputs.	Passed - the training state and available actions were preserved when returning to the Model/Training stage.
CKPT-01	Detect available checkpoints	At least one checkpoint exists	Open checkpoint selection area	System lists available checkpoints.	Passed - available checkpoints were detected and listed.
CKPT-02	Display checkpoint metadata	Checkpoint has config metadata	Inspect checkpoint	System displays run name, model settings, and training configuration.	Passed - checkpoint metadata and configuration information were displayed.
CKPT-03	Prevent evaluation without selected checkpoint	No checkpoint selected or file missing	Attempt evaluation	System prevents evaluation and displays a checkpoint error message.	Passed - the Evaluation workflow required a valid selected checkpoint before evaluation could proceed.
CKPT-04	Use selected checkpoint for evaluation	Valid checkpoint exists	Select checkpoint and continue	Selected checkpoint is stored and used by evaluation stage.	Passed - the selected checkpoint was stored and successfully used by the Evaluation stage.

2.6 Evaluation Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
EVAL-01	Open Evaluation stage with valid checkpoint	Valid checkpoint selected	Navigate to Evaluation	System displays evaluation configuration and allows running evaluation.	Passed - the Evaluation stage opened with the selected checkpoint and displayed evaluation configuration.
EVAL-02	Prevent evaluation without checkpoint	No valid checkpoint selected	Attempt evaluation	System prevents evaluation and displays clear message.	Passed - evaluation readiness required a valid selected checkpoint before evaluation execution.
EVAL-03	Run evaluation	Valid dataset, embeddings, and checkpoint exist	Start evaluation	Evaluation process runs and displays progress/status updates.	Passed - evaluation ran successfully using valid dataset artifacts, embeddings, and checkpoint.
EVAL-04	Compute evaluation metrics	Evaluation completes successfully	Inspect evaluation output	System computes and displays metrics such as MAE and sample count.	Passed - evaluation metrics, including MAE/MSE and per-dimension values, were computed and displayed.
EVAL-05	Save prediction examples	Evaluation configured to save examples	Run evaluation	Prediction examples are saved and available in Results screen.	Passed - prediction examples were saved and became available for Results review.
EVAL-06	Evaluation log generation	Evaluation running or completed	Open evaluation log	Log contains model loading, dataset loading, metrics, and completion messages.	Passed - evaluation logs were generated and displayed with model/data loading and completion messages.
EVAL-07	Handle failed evaluation	Missing files or incompatible configuration	Run under invalid conditions	System reports failure clearly and does not display misleading results.	Passed - evaluation input validation and artifact guards prevented misleading evaluation results.
EVAL-08	Re-run evaluation	Previous evaluation exists	Run evaluation again	System updates output and displays latest valid results.	Passed - re-running evaluation updated the output/results workflow successfully.

2.7 Results Visualization Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
RES-01	Open Results stage after evaluation	Evaluation completed successfully	Navigate to Results	System displays evaluation results.	Passed - the Results stage opened and displayed the evaluation output.

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
RES-02	Display metrics	Evaluation metrics exist	Open Results screen	Metrics such as MAE and sample count are displayed clearly.	Passed - metric cards were displayed clearly.
RES-03	Display prediction charts	Prediction and ground truth values exist	Inspect charts area	Charts display predictions and ground truth without being cut off or misaligned.	Passed - prediction-vs-ground-truth charts were displayed correctly.
RES-04	Enable chart interaction	Charts displayed	Use mouse zoom or pan	Chart supports expected interaction and remains visually stable.	Passed - chart zoom/pan interaction and visual stability were validated.
RES-05	Display prediction example image	Examples include image/frame info	Open example viewer	Corresponding image or valid placeholder is displayed.	Passed - prediction example image display was validated.
RES-06	Navigate prediction examples	Multiple examples exist	Click First/Previous/Next/Last	Displayed example changes correctly and counter/progress updates.	Passed - navigation between prediction examples updated the example, counter, charts, and table.
RES-07	Display action comparison table	Prediction and ground-truth vectors exist	Inspect table	Table displays dimensions, predictions, ground truth, and absolute error.	Passed - the action comparison table displayed dimensions, predictions, ground truth, and error values.
RES-08	Results layout consistency	Results include metrics, charts, images, controls, table, log	Inspect full screen	Elements remain within frames, aligned, readable, and not cut off.	Passed - the Results layout remained readable, aligned, and visually consistent.
RES-09	Display evaluation log	Evaluation log exists	Inspect log area	Log appears in correct section and does not overlap other content.	Passed - the evaluation log was displayed in the correct Results section without overlapping content.

2.8 General Flow, State Persistence, and Error Handling Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
FLOW-01	Pipeline navigation	UI is open	Move between stages	Active stage is highlighted and only valid stages are accessible.	Passed - pipeline navigation behaved correctly and active stages were highlighted.
FLOW-02	Prevent progress during running process	A stage is running	Try to continue	System prevents progress until process completes or stops.	Passed - progress to dependent stages was blocked while long-running stages were active.
FLOW-03	Back navigation after completed stage	At least one stage completed	Navigate back	Stage displays correct completed/ready status and retains artifacts.	Passed - completed and ready states were preserved during back navigation.
FLOW-04	State restoration after reopening UI	Stage completed/stopped/partial/ready	Close and reopen UI	System restores correct state based on files, logs, and process status.	Passed - UI state restoration after reopening was validated using project artifacts and process state.
FLOW-05	New dataset resets dependent stages	Dataset already selected	Select new dataset	Dependent stages reset or update as needed.	Passed - dependent stages reset or updated correctly when the selected dataset changed.
FLOW-06	Logs separated by stage	Logs exist for multiple stages	Move between screens	Each screen displays only relevant logs.	Passed - stage-specific logs were displayed consistently without mixing unrelated stage logs.
FLOW-07	Missing artifact handling	Required artifact missing	Try dependent operation	System displays clear error and prevents invalid operation.	Passed - missing/invalid artifact handling was validated through UI guards and invalid dataset handling.
FLOW-08	UI layout consistency	UI contains buttons, charts, tables, images, logs	Inspect screens after pipeline run	Layout remains readable, aligned, and not cut off.	Passed - main UI screens remained readable and aligned after running the pipeline.
FLOW-09	Stop and resume consistency	Long process was stopped safely	Resume process	Process continues from saved partial state unless restart explicitly requested.	Passed - stop/resume behavior continued from saved partial state for long-running workflows.

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
FLOW-10	Final end-to-end workflow	Valid dataset, models, and environment available	Run workflow from Dataset to Results	System completes pipeline and displays metrics, charts, examples, tables, and logs.	Passed - the complete workflow from dataset selection to final results visualization was validated.

2.9 Data Consistency Validation Tests

Test Case ID	Description	Preconditions	Test Steps	Expected Result	Actual Result
DATA-CONS-01	Validate state/action length relation	Trajectory has observations and actions	Compare counts	System confirms expected relation such as N observations and N-1 actions.	Passed - state/action sequence relation supported sample construction during data loading.
DATA-CONS-02	Validate action vector size	Policy output files available	Load action vectors	Each action vector has expected dimensionality for prediction.	Passed - action vectors were loaded and used successfully for 7-dimensional action prediction.
DATA-CONS-03	Validate state vector size	Observation files available	Load state vectors	Each state vector has expected dimensionality and can be used as input.	Passed - robot-state vectors were loaded and used successfully as model input.
DATA-CONS-04	Validate gripper-related component	State, desired state, and action vectors available	Compare last component over time	Last component behaves consistently with gripper open/close behavior.	Passed - gripper-related information was handled consistently during filtering, sequence construction, training, and evaluation.
DATA-CONS-05	Validate action-to-next-state alignment	State, desired state, and action sequences available	Compare action at t with target at following timestep	Selected alignment is consistent with behavioral cloning configuration.	Passed - behavioral cloning samples were constructed with target alignment consistent with the configured strategy.
DATA-CONS-06	Validate timestamp consistency	Timestamp data exists	Inspect ordering and gaps	Timestamps are ordered and do not contain invalid jumps.	Passed - ordered sequence data supported valid temporal window construction.
DATA-CONS-07	Validate qpos/qvel consistency	Joint position and velocity data exist	Compare qpos changes with qvel	qvel behaves consistently with joint motion without invalid unexplained values.	Passed - robot joint/state data was loaded and used consistently by the model input pipeline.
DATA-CONS-08	Validate end-effector pose data	eef_transform data exists	Inspect transform matrix dimensions and values	End-effector transform has expected structure and can be used for validation when needed.	Passed - available trajectory pose/transform fields were handled consistently and did not block sequence construction, training, or evaluation.
DATA-CONS-09	Validate filtered success trajectories	Filtered CSV exists	Count valid/SUCCESS trajectories	Filtered dataset contains expected usable trajectories for selected configuration.	Passed - filtered trajectory status was consistent with the selected training/evaluation workflow.
DATA-CONS-10	Validate training window count	Dataset loading completed	Compare reported windows with configuration	Generated windows are consistent with valid trajectories, sequence length, and filtering.	Passed - generated samples/windows were reported consistently with valid trajectories, sequence length, and filtering configuration.